

FIT5202 – Data Processing for Big Data

Granularity Reduction and Sensor Arrays



Last Week

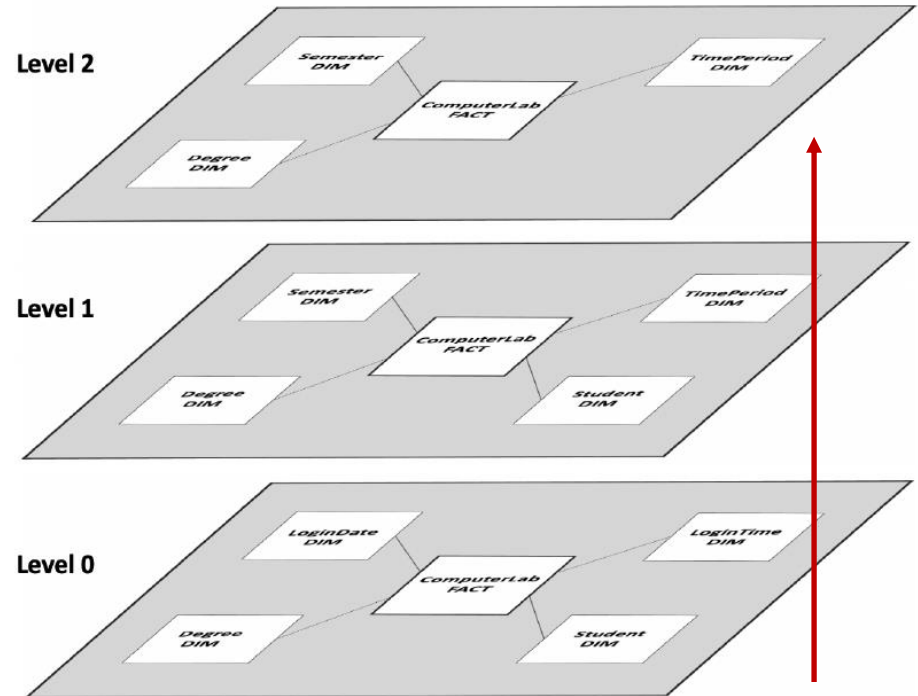
- Overview of Stream join
- Time based window stream join (Unbounded)
- Tuple based window stream join (Unbounded)
- Bounded stream join

This Week

- Granularity Reduction in Data Streams
- Mixed Levels of Granularity
 - How to combine two data streams with different levels of granularity
- Sensor Arrays
 - How to process data streams collected from an array of sensors

Granularity

- **Granularity** is the **level of detail** at which **data** are stored in a database.
- level-0, the bottom level indicating no aggregation (raw data) → **high granularity**.
- level-1 and level-2 with more aggregation → **lower granularity**.
- **Motivation** of having multiple levels: Direct querying on higher level of aggregation is more efficient, considering the high volume & high velocity of data stream



Details in data
reduced at
higher levels

Granularity Reduction in Data Streams

- Granularity is not only for efficient retrieval.
- Granularity is also about **managing complexity**.
 - higher granularity → raw data, complete but complex
 - lower granularity → data simplified (or summarized), less complex & less details

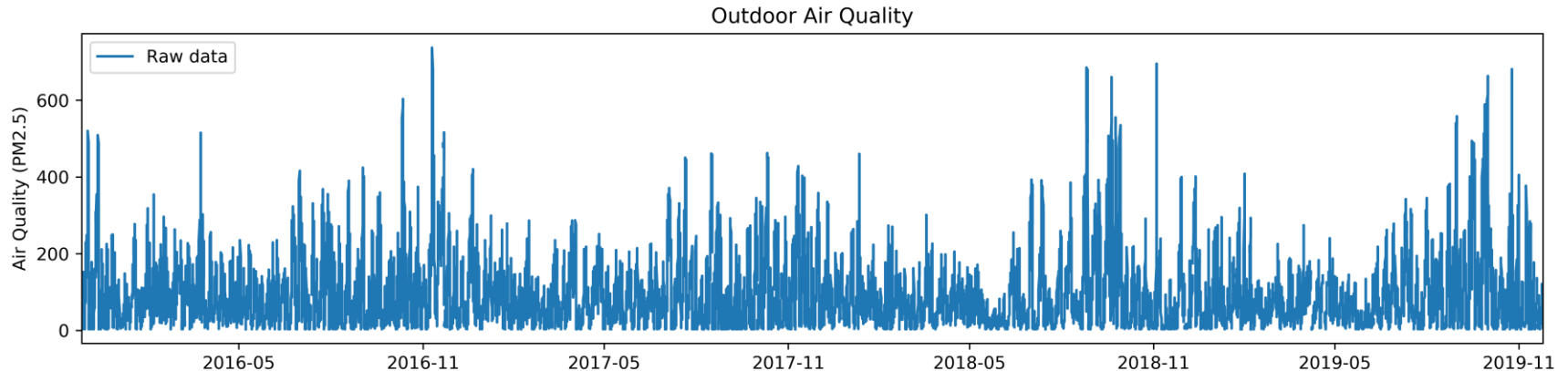


Figure1: Outdoor Air Quality of a city, measured in PM2.5 (raw data collected over hours)

Granularity Reduction in Data Streams

- Granularity is not only for efficient retrieval.
- Granularity is also about managing complexity.

To reduce granularity: Use mean function to **aggregate hourly data into one day data**

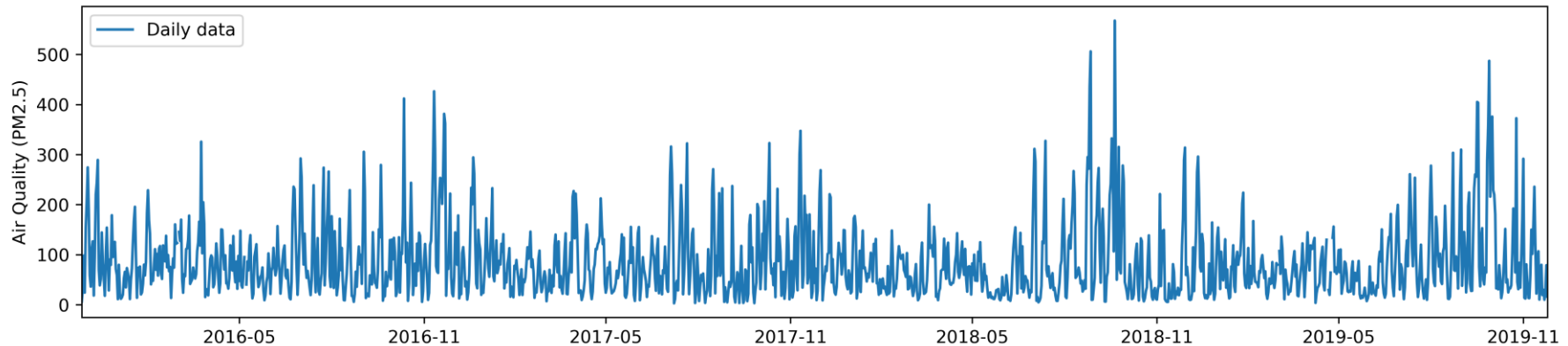


Figure2: Outdoor Air Quality of a city, measured in PM2.5 (Daily Aggregation)

Granularity Reduction in Data Streams

- Granularity is not only for efficient retrieval.
- Granularity is also about managing complexity.

Weekly aggregation – some trend is observed

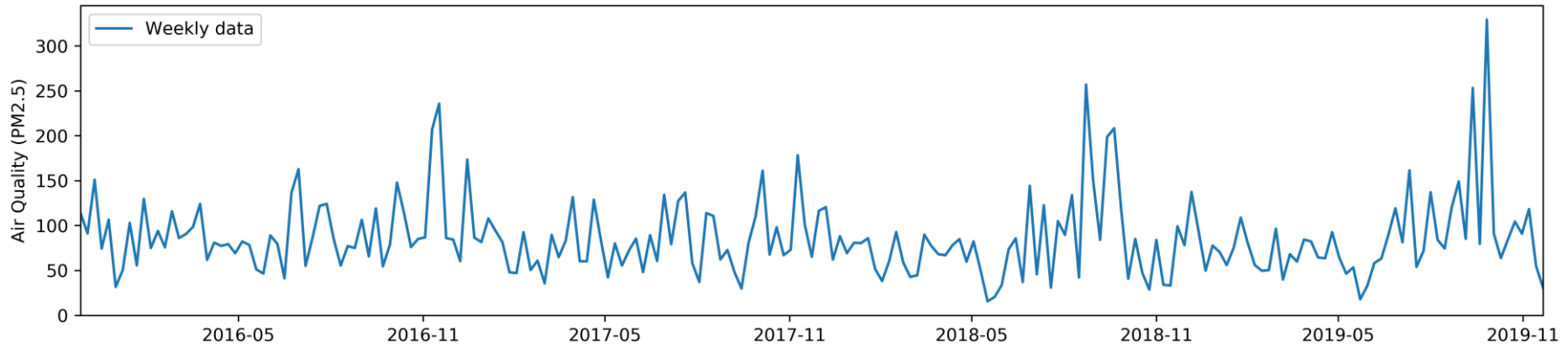


Figure3: Outdoor Air Quality of a city, measured in PM2.5 (Weekly Aggregation)

Granularity Reduction in Data Streams

- Granularity is not only for efficient retrieval.
- Granularity is also about managing complexity.
- Granularity may simplify the complexity of the information.

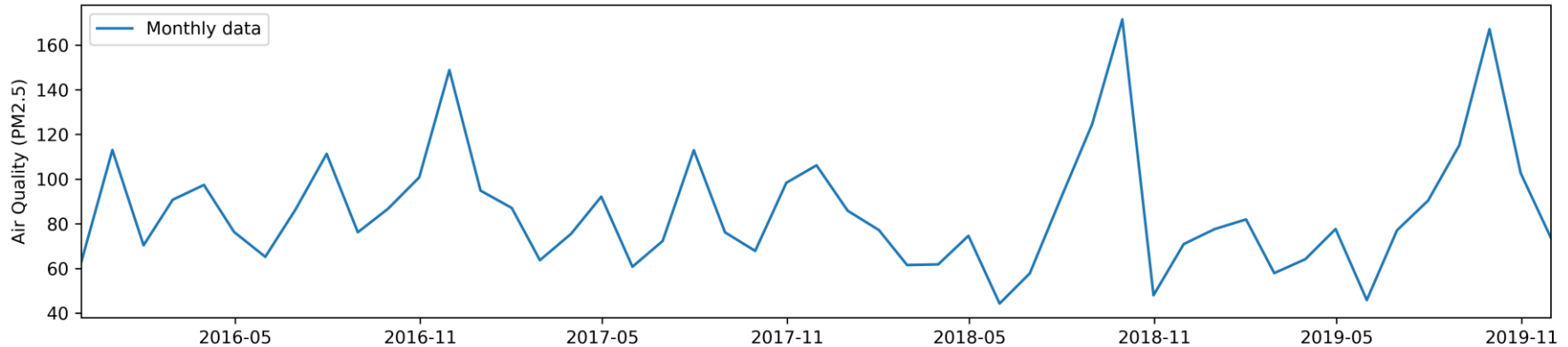


Figure4: Outdoor Air Quality of a city, measured in PM2.5 (Monthly Aggregation)

Fixed-Size Windows

- **Time-based windowing** is used to reduce granularity in data streams
 - Window size is uniform throughout the data stream.
 - Time-slide is applied at regular interval.
- Two time-slide approaches for fixed-size time-based windows:
- **Overlapped Windows**
 - Slide time is less than the window size
- **Non-overlapped Windows**
 - Slide time is equivalent to the window size

Fixed-Size Windows

- **Overlapped Windows:** No Granularity Reduction
 - When windowing is applied, normally, the aim is to calculate the mean (or any other statistical function) of values within the window.
 - For **time-slide of 1 time unit/1 record** → **No reduction in terms of number of records.** This is a pure moving average (also know as rolling mean).

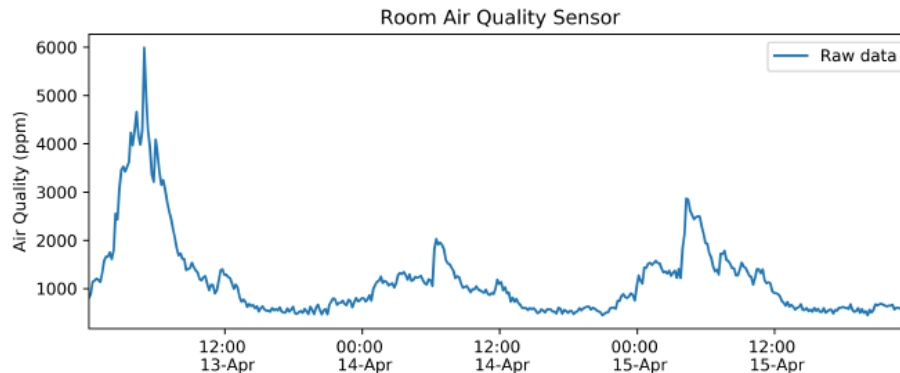
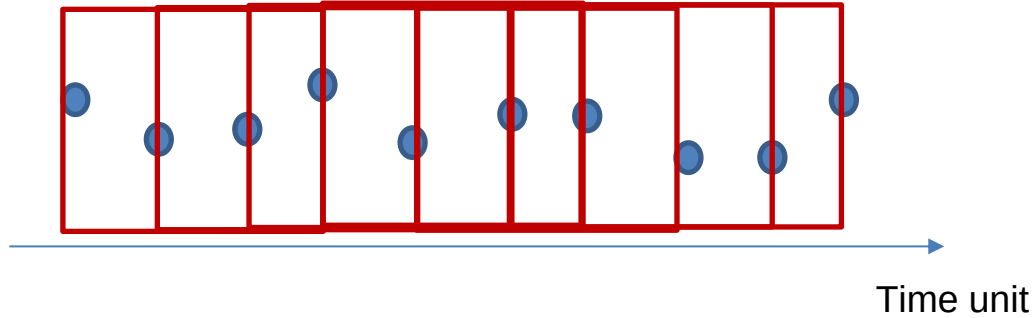


Figure5: The Indoor Air Quality for 3 days (10 mins sensor recording)

Overlapped Windows: No Granularity Reduction



Window size = 4 unit of time / 4 record
Slide = 1 unit time / 1 record



Averaging within
window



Fixed-Size Windows

■ Overlapped Windows: No Granularity Reduction

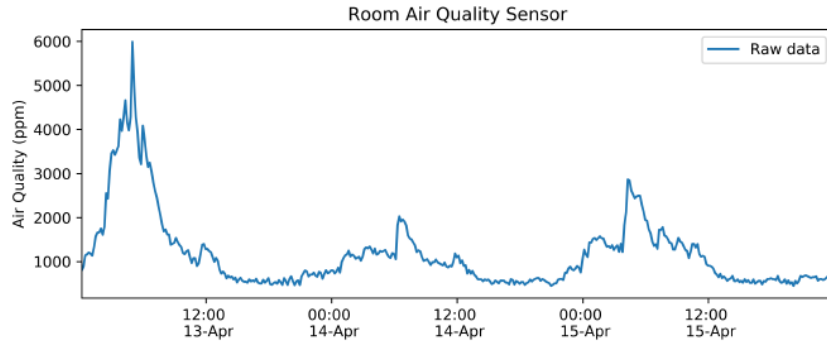


Figure5: The Indoor Air Quality for 3 days (10 mins sensor recording)

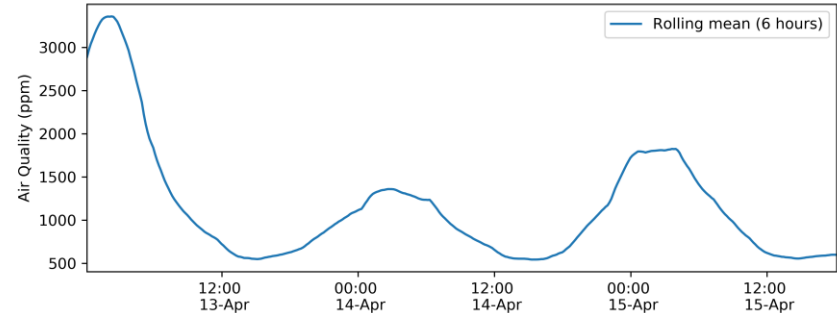


Figure6: Rolling mean of 6 hours (window size = 6 hours, time-slide every record/ 10 mins).

- Number of data points is not reduced
- Rolling mean (moving average) smooth out data points – revealing smoother trend
- The longer the window size – the smoother the trend

Fixed-Size Windows

- **Overlapped Windows:** With Granularity Reduction
 - When the **time slide is more than one unit of time**, there will be a reduction in terms of number of records after aggregation.

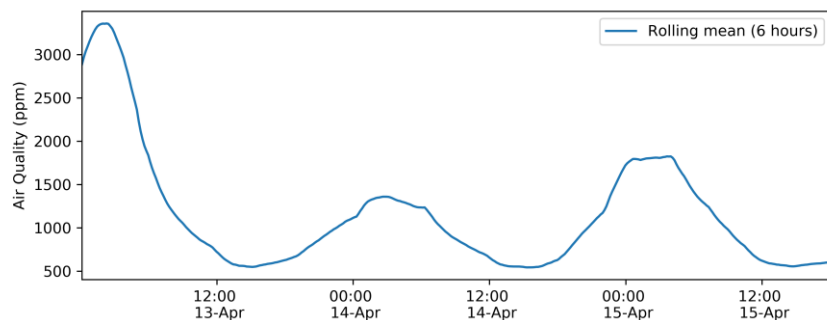


Figure6: Rolling mean of 6 hours.
(window size = 6 hours, time-slide = 10 mins)

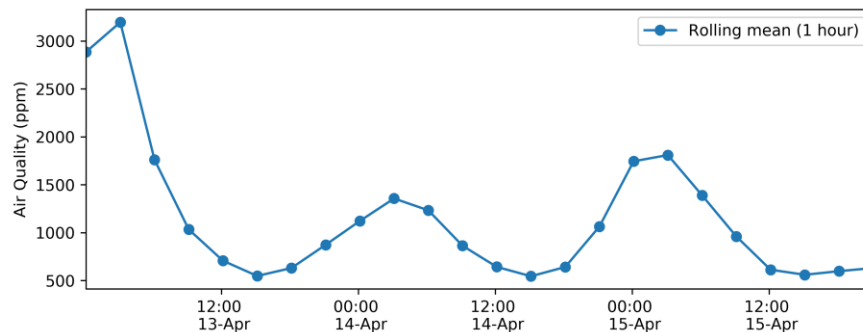
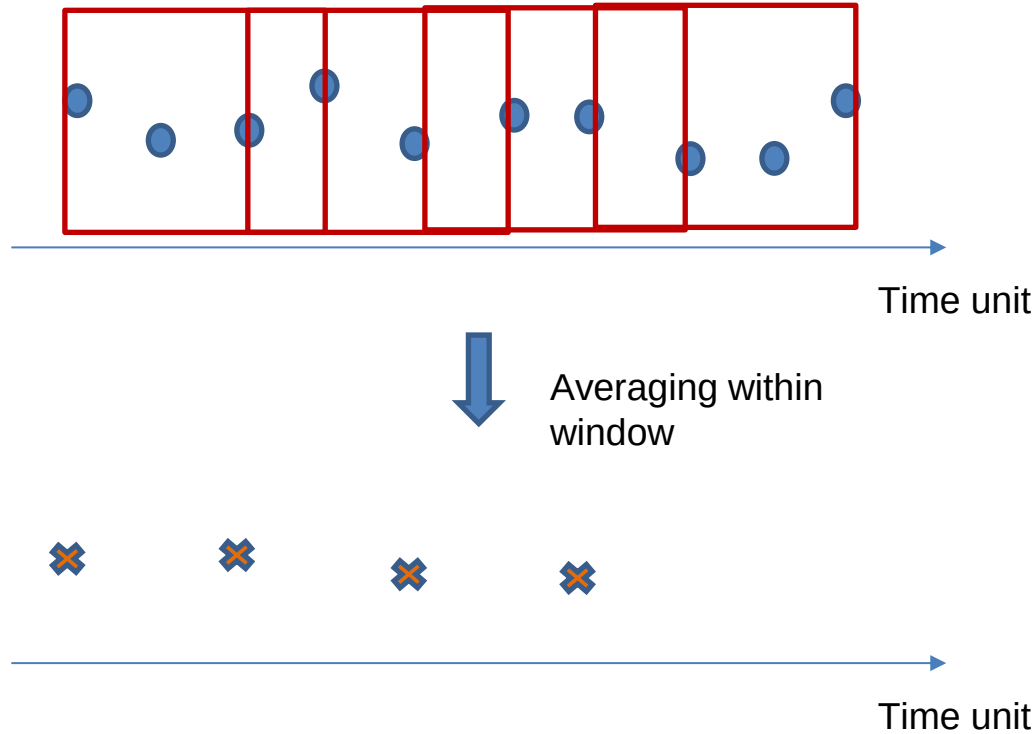


Figure7: Window size = 6 hours, time-slide = 3 hours.

→ Data will be produced every 3 hours, instead of every 10 mins

Overlapped Windows: Granularity Reduction



Window size = 4 unit of time/ 4 record
Slide = 2 unit time / 2 record

Fixed-Size Windows

- **Non-Overlapped Windows:** Granularity Reduction
 - Consecutive windows are not overlapped, but there is no gap between the windows.

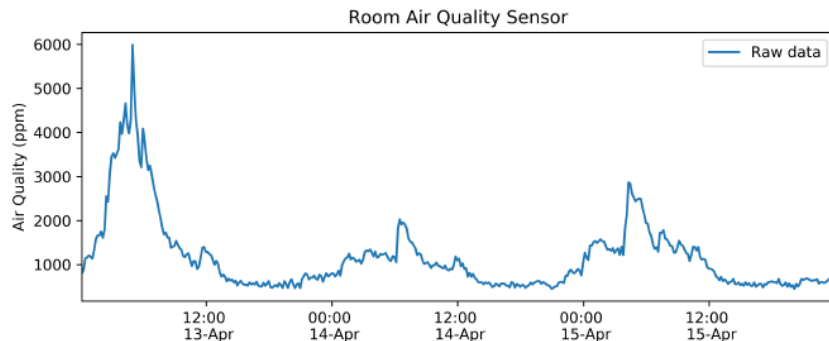


Figure5: The Indoor Air Quality for 3 days

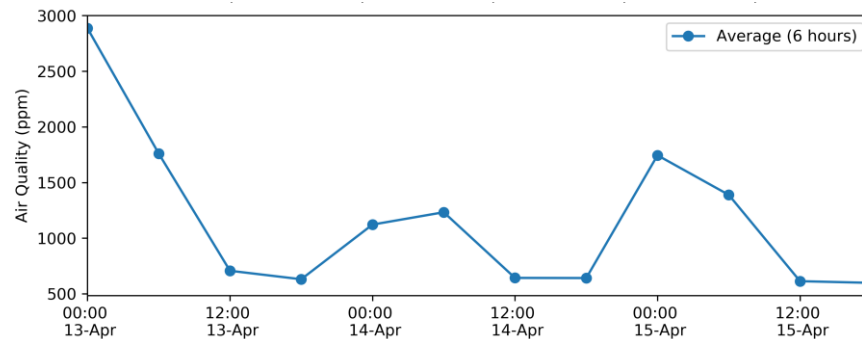
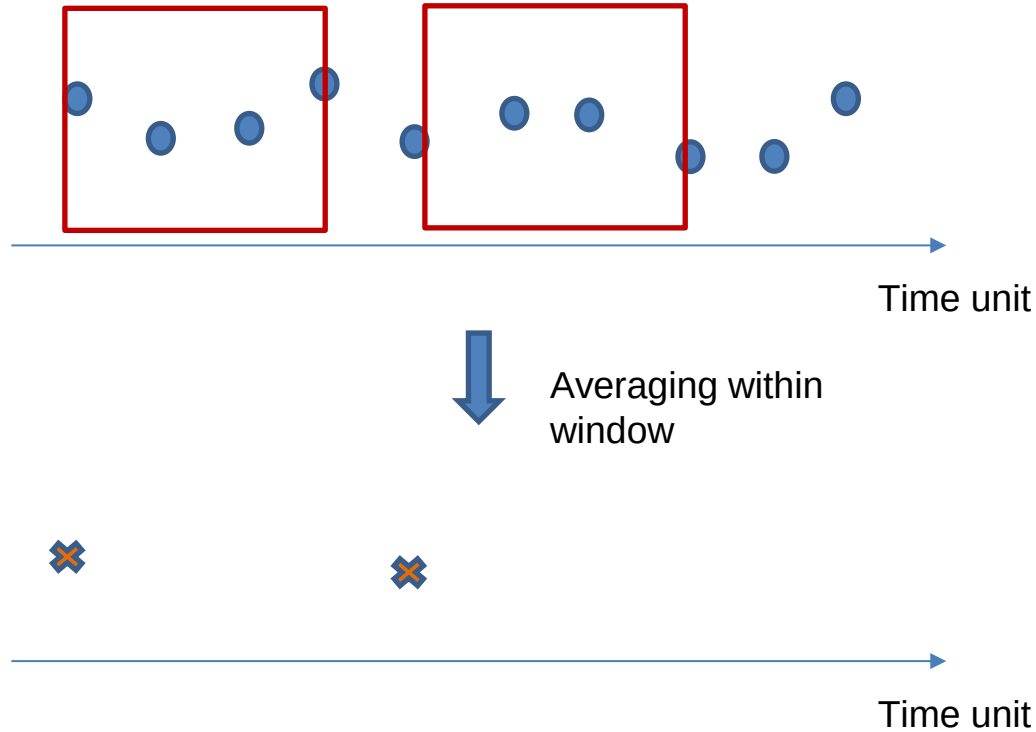


Figure8: The window size of 6 hours (time slide = 6 hours), and consecutive windows are not overlapped.

Besides averaging, the reduction function in the window can be any aggregate reduction function, such as average, min or max.

Non-Overlapped Windows: Granularity Reduction



Window size = 4 unit of time / 4 record
Slide = 4 unit time / 4 record

Mixed Levels of Granularity

- Different levels of granularity combined into one level.
- Mixed level of granularity can be two types:
 - Temporal-based Mixed Levels of Granularity
 - Time based.
 - Spatial-based Mixed Levels of Granularity
 - Space or location based.

Mixed Levels of Granularity

- Temporal-based Mixed Levels of Granularity
 - Time based.

A typical granularity reduction is shown in Figure 9 (top) and 10 (bottom), which are hourly granularity and 6-hourly granularity. These use fixed size windows.

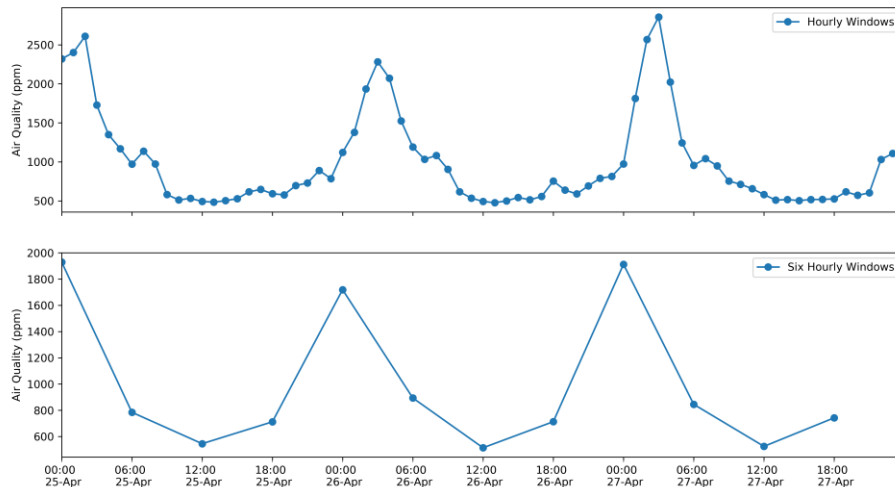
1-hour window granularity:

- win size = 1 hour, slide = 1 hour

6-hour window granularity:

- win size = 6 hours, slide = 6 hours

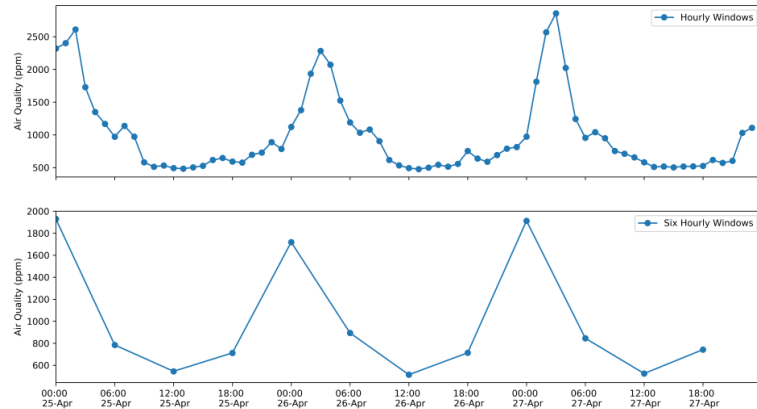
Reveal seasonal trend



- hourly granularity has more details than the 6-hourly granularity

Mixed Levels of Granularity

- Temporal-based Mixed Levels of Granularity
 - Time based.



Mixing

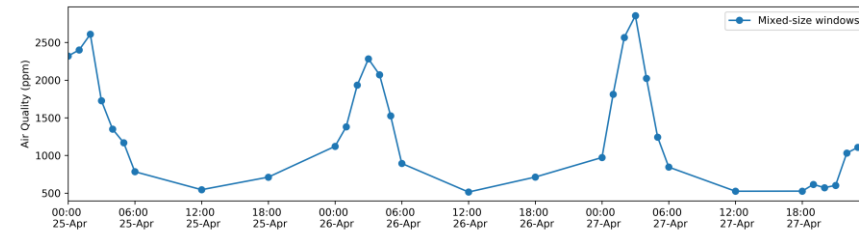


Figure11: mixing hourly granularity and 6-hourly gran
These use fixed size windows.

- ❑ Combine different levels of granularity
- ❑ It captures generality of 6 hour windows
- ❑ At certain period, granularity is further broken down into lower level

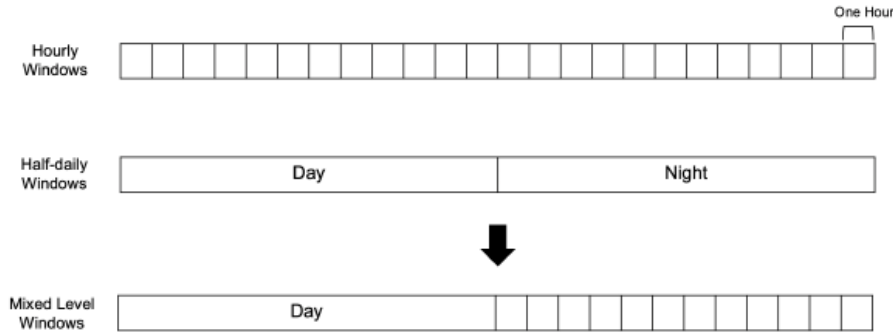


“Drill-down”

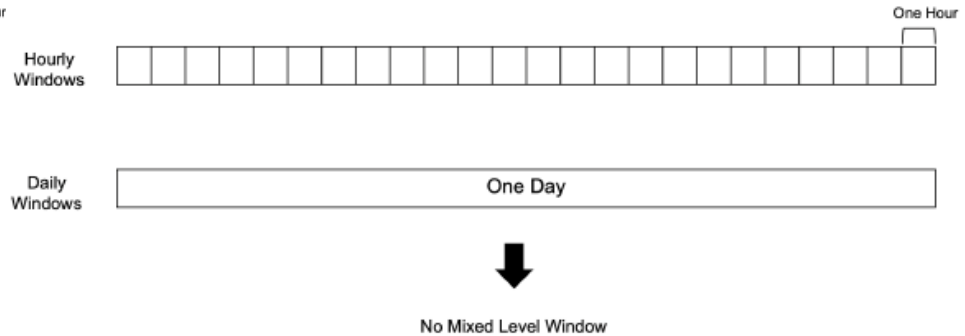
- Allow users to move from summary/aggregate information to detailed data

Mixed Levels of Granularity

- Temporal-based Mixed Levels of Granularity
 - Time based.



(a) Mixed Level of Granularity between Day and Night



(b) No Mixed Level

Require categorization of data

Mixed Levels of Granularity

- Spatial-based Mixed Levels of Granularity
 - Location based.

Figure 12: The tram data based on Stop Number

- Level-0 granularity

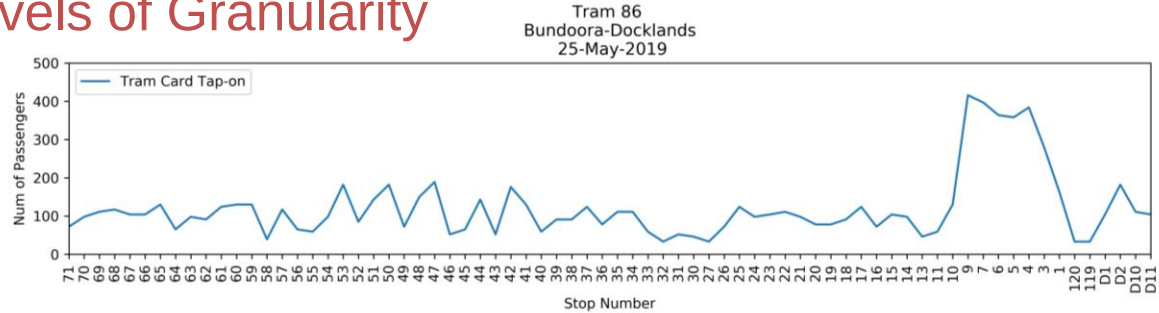
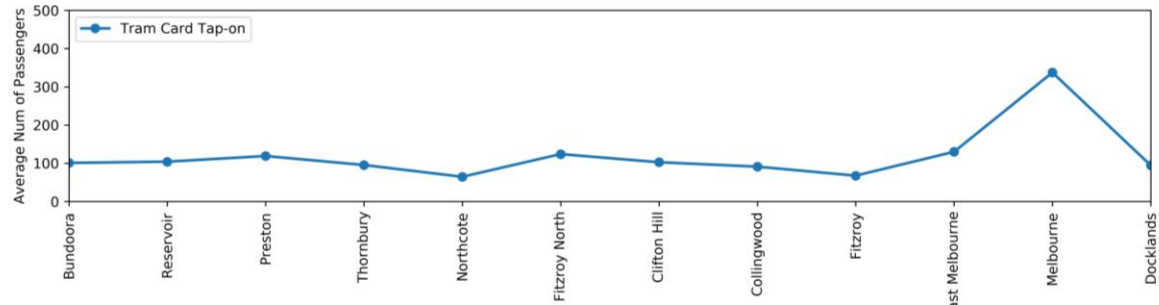


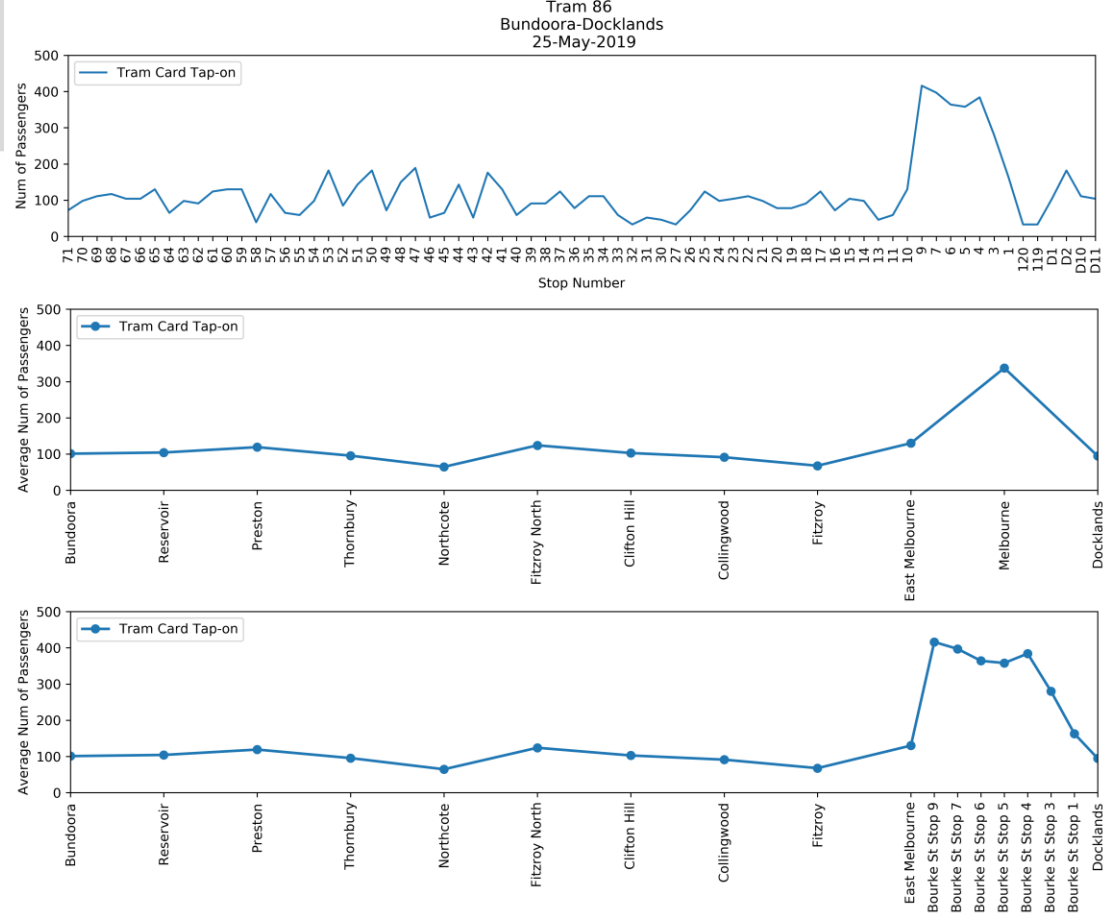
Figure 13: The tram data based on Suburb (average number of passengers for each suburb)

- Level-1 granularity



Mixed Levels of Granularity

- Spatial-based Mixed Levels of Granularity - Space based.
- Ex: drill-down is applied to the suburb with high intake of passenger, - Melbourne CBD
- Data in Melbourne CBD is expanded to the stop level to reveal details of each stop.



Sensor Arrays

- A sensor array is a **group of sensors, usually deployed in a certain geometry pattern.**
- A network of distributed sensors.
- **Advantage:** add new dimensions to the observation, and hence it helps to estimate more parameters, to have a better picture of the environment being observed, and to improve accuracy.
- Two categories:
 1. **Multiple sensors measure the same things**, and
 2. **Multiple sensors measure different things**, but they are grouped together.

Sensor Arrays

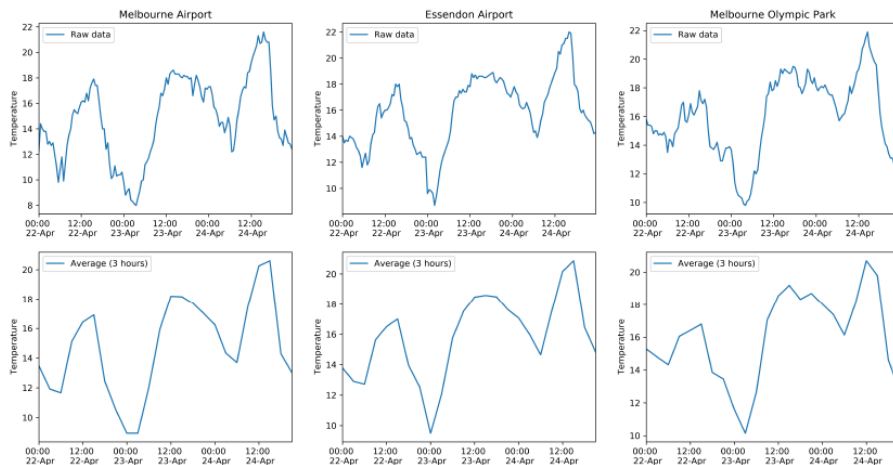
- Multiple sensors measuring the same things
- **Why?**
 - Specialize on sensing a very specific small region
 - Get more accuracy of the results for measuring the entire region or condition
 - Failsafe/Redundancy
- **Example:** Multiple weather stations in a region or city measure the air temperatures. We use three weather stations in Melbourne: Melbourne Airport, Melbourne Olympic Park and Essendon Airport.
- The aggregate of these three weather stations represents the temperature of Melbourne city. So these three weather stations are measuring the same thing i.e. the air temperature

Sensor Arrays

- Multiple sensors measuring the same things
- Two methods to lower the granularity of sensor arrays that measure the same thing:
 - Method 1: Reduce and then Merge
 - Method 2: Merge and then Reduce

Sensor Arrays

- Multiple sensors measuring the same things
- **Method 1: Reduce and then Merge**
 - Step 1: Reduce the granularity of each individual sensor.

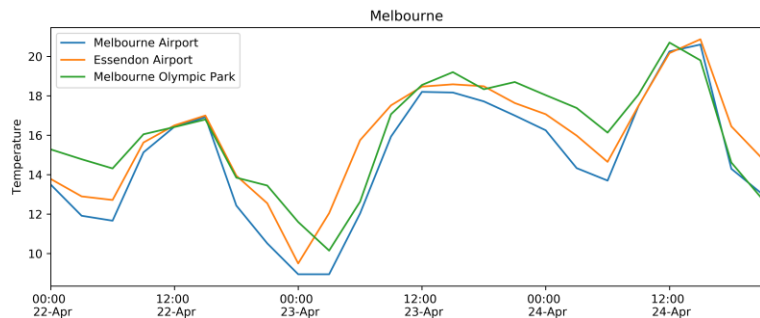


- Raw data – half-hourly data

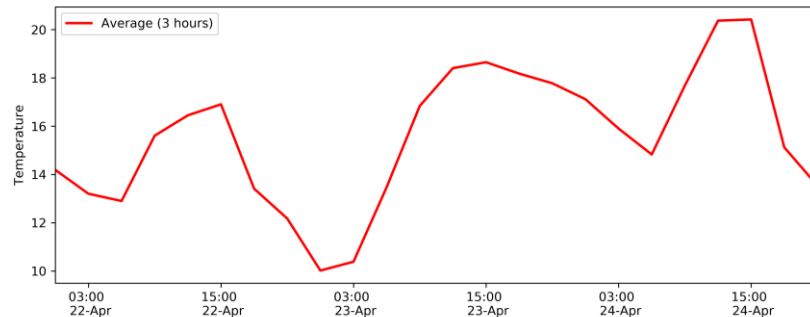
- Average over 3 hour data

Sensor Arrays

- Multiple sensors measuring the same things
- **Method 1: Reduce and then Merge**
 - Step 2: Merge reduced granularity into one data stream.



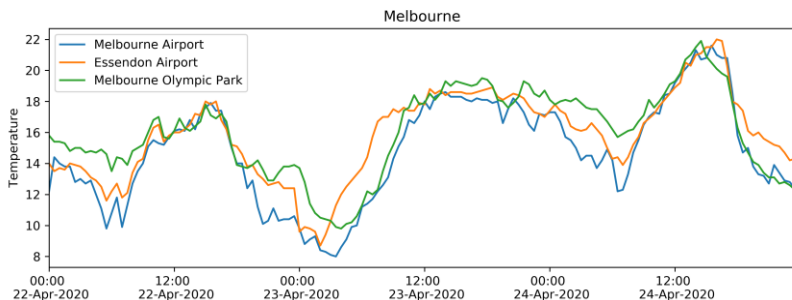
Merge



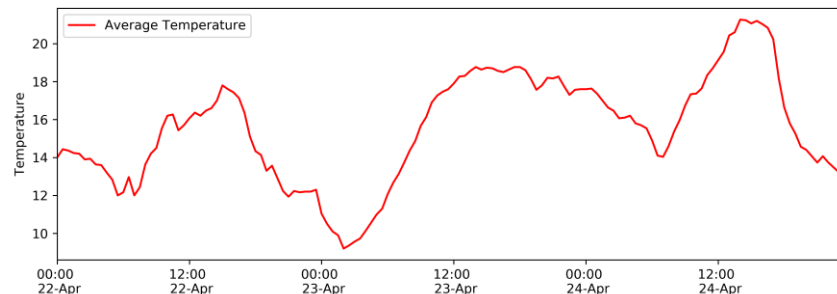
Merging: aggregate three weather stations at the 3-hour granularity, based on timestamp (e.g., using mean function)

Sensor Arrays

- Multiple sensors measuring the same things
- **Method 2: Merge and then Reduce**
 - Step 1: Merge the data streams first.



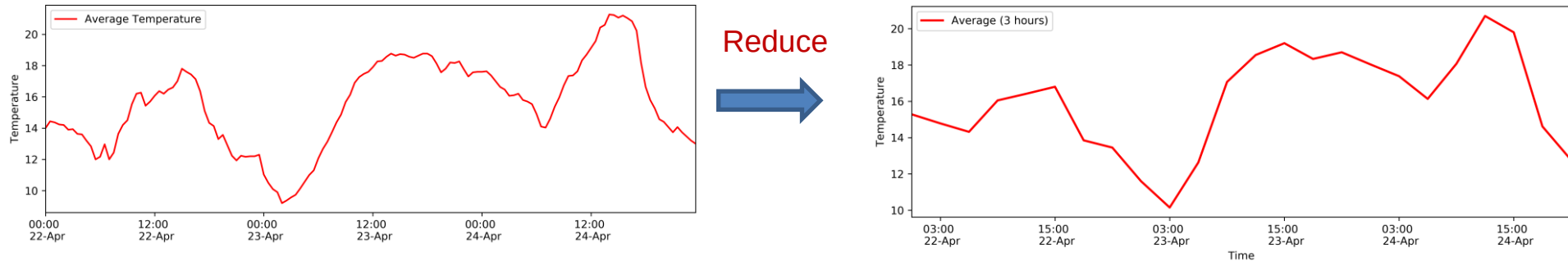
Merge



Merging: aggregate the raw temperature data over three weather stations

Sensor Arrays

- Multiple sensors measuring the same things
- **Method 2: Merge and then Reduce**
 - Step 2: Reduce the **granularity** of the merged data stream.



Both methods shows similar trend. For Method 2: some of details in the raw data are still preserved in the reduced granularity version

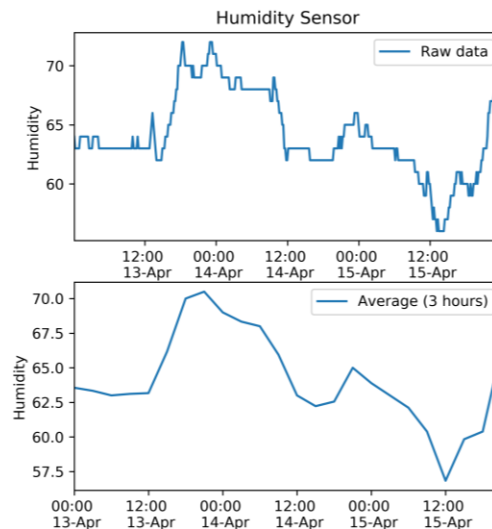
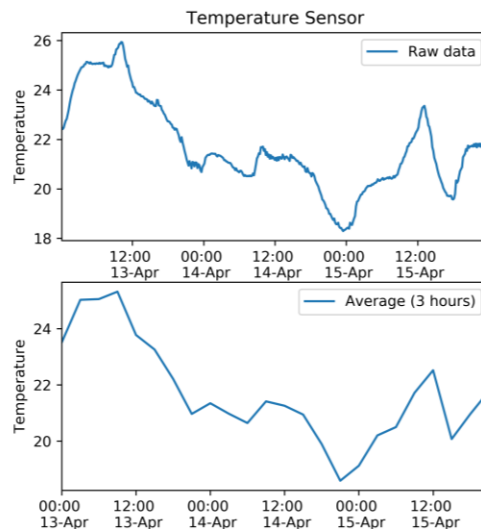
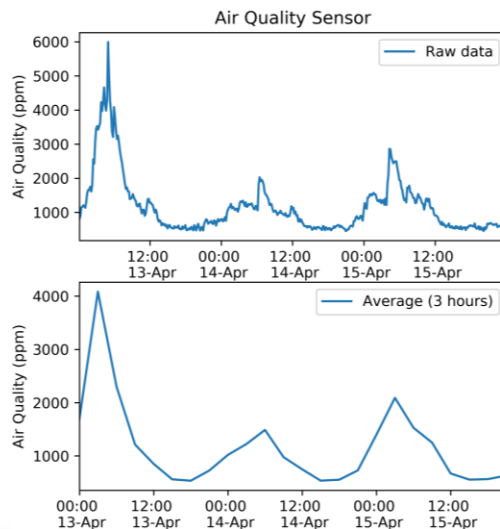
→ **Reason:** In Method 1, details of raw data have been smoothed out in first step. In Method 2, reduced step is still based on level-0 (raw) data

Sensor Arrays

- **Multiple sensors measuring different things**
 - Sensor arrays can be a collection of sensors measuring different things within the same environment.
- Example: A simple indoor sensor array, containing three sensors: air quality, temperature, and humidity -> Level of Comfort
- **Two methods to lower the granularity of sensor arrays that measure the different things:**
 - Method 1: Reduce, Normalize, and then Merge
 - Method 2: Normalize, Merge and then Reduce

Sensor Arrays

- Multiple sensors measuring different things
- **Method 1: Reduce, Normalize, and then Merge**
 - Step 1: Reduce the granularity level of each sensor's raw data.

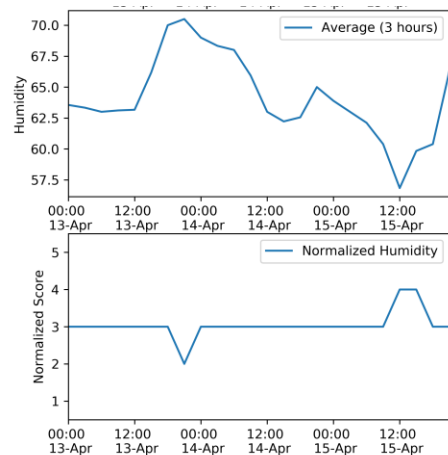
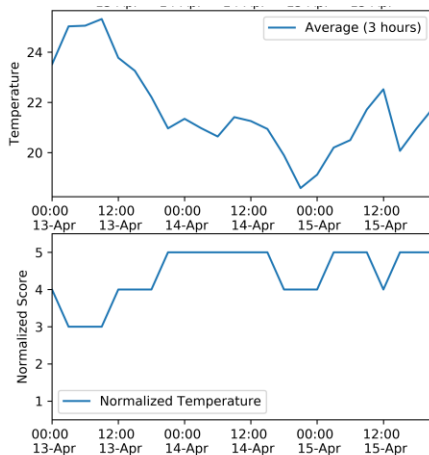
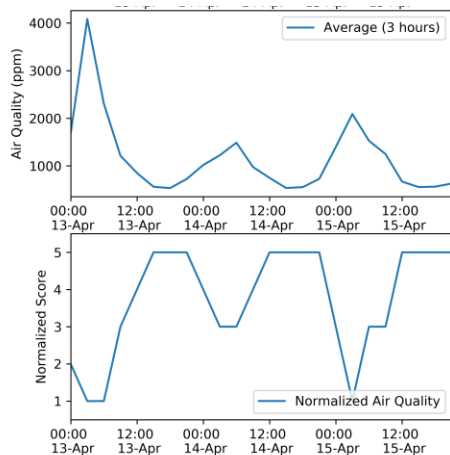


Sensor Arrays

- Multiple sensors measuring different things
- **Method 1: Reduce, Normalize, and then Merge**

Why do we need normalization?

- Step 2: Normalize the reduced data of each sensor by categorizing each data into several categories (e.g., category 1 – Poor, category 5 – excellent)

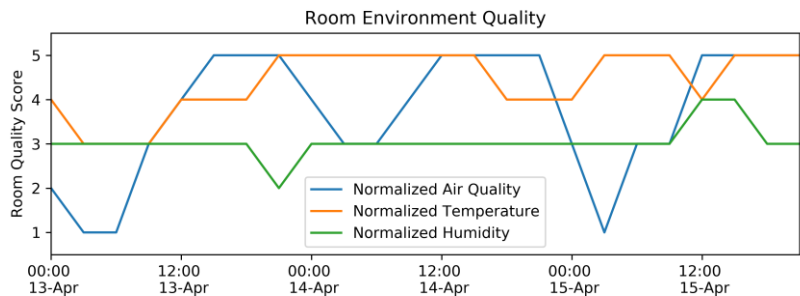


Ex: 5 level of categories of room air quality:

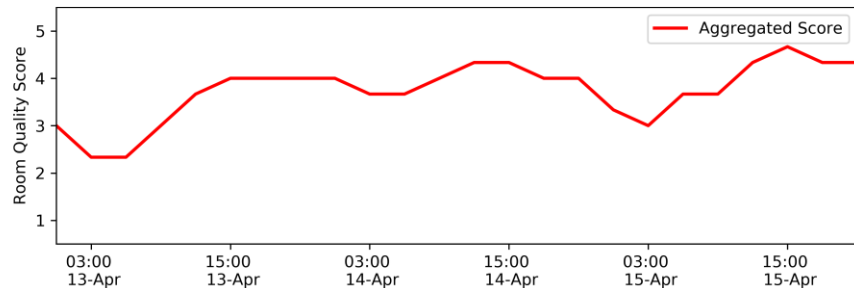
1. Excellent: < 800 ppm
2. Good: 800 to < 1100 ppm
3. Fair: 1100 to < 1600 ppm
4. Inferior: 1600 to < 2000 ppm
5. Poor: 2000 ppm or over

Sensor Arrays

- Multiple sensors measuring different things
- **Method 1: Reduce, Normalize, and then Merge**
 - Step 3: Merge the normalized data.

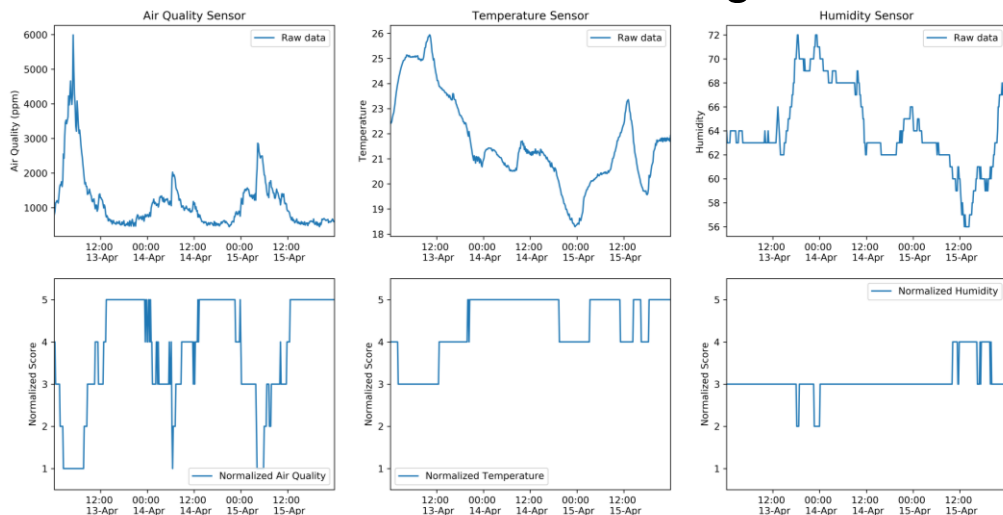


Merge



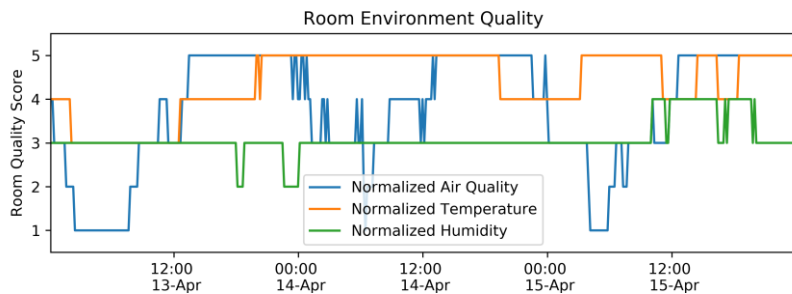
Sensor Arrays

- Multiple sensors measuring different things
- Method 2: Normalize, Merge and then Reduce
 - Step 1: Normalize each stream according to the categories.

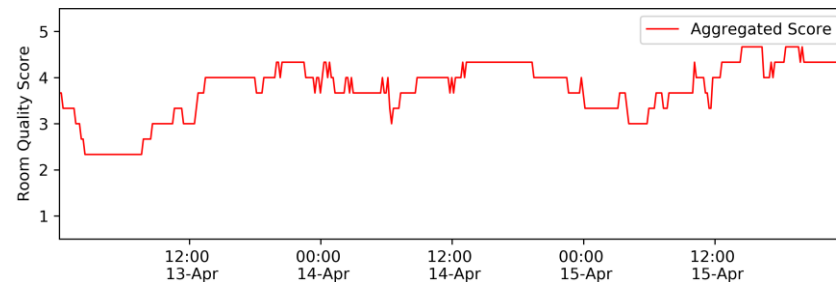


Sensor Arrays

- Multiple sensors measuring different things
- **Method 2: Normalize, Merge and then Reduce**
 - Step 2: Merge the normalized streams.

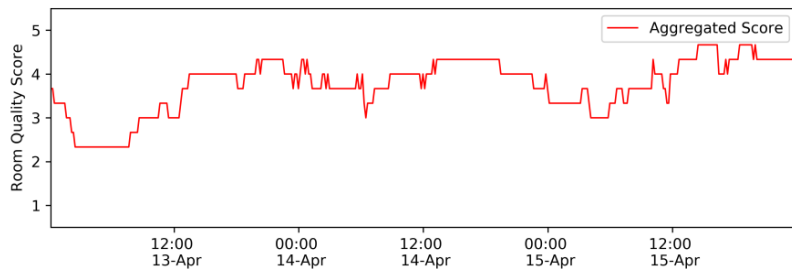


Merge

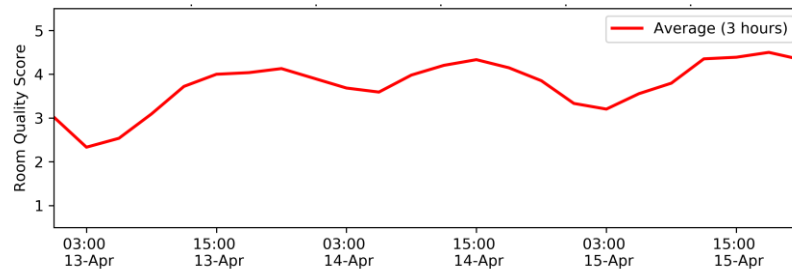


Sensor Arrays

- Multiple sensors measuring different things
- **Method 2: Normalize, Merge and then Reduce**
 - Step 3: Reduce the granularity of the merged results.



Reduce



Half-hourly data data is reduced to 3-hourly data

Summary

- Granularity reduction of sensor data is achieved through the **windowing schemes**.
- The **drill-down of data streams** can be assisted through multi levels of granularity which combined several granularity levels when presenting the data streams.
- Sensor arrays are multiple sensors that work together in an environment to provide users with more complete picture of the environment.